

Estimated Annual Agricultural Pesticide Use Adams County, Washington 2014-2018

24 July 2026

Summary

Adams County ranks **#4 of Washington’s 39 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of WA’s five highest pesticide-use counties. Adams County is Columbia Basin wheat country, farmed under both dryland and irrigated systems, alongside potatoes and other irrigated row crops.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Adams County, Washington** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound’s chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” -[Pesticide National Synthesis Project webpage](#)

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”).
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
OTH	Other
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Adams County, WA, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	METAM	DTC	1457865	1428457	1133223
2	DICHLOROPROPENE	OC	961753	985098	932557
3	METAM POTASSIUM	DTC	392225	257672	264637
4	CHLOROPICRIN	OC	134727	58619	38585
5	SULFURIC ACID	INO	94991	208553	224700
6	GLYPHOSATE	OTH	70203	94352	86057
7	PETROLEUM OIL	OTH	69034	95154	93156
8	BACILLUS AMYLOLIQUEFACIEN	MIC	41755	44305	31814
9	2,4-D	CPX	37473	38569	37654
10	CALCIUM POLYSULFIDE	INO	32979	20302	17394
11	SULFUR	INO	32328	35393	38421
12	EPTC	TC	26346	30235	39296
13	MANCOZEB	DTC	24531	30925	38678
14	CHLOROTHALONIL	OTH	17753	17950	18375
15	OXAMYL	CB	10804	4336	8247
16	PARAQUAT	DIP	8913	11569	9599
17	PROPICONAZOLE	OTH	8292	8488	6581
18	METRIBUZIN	TRZ	7945	7420	7528
19	BROMOXYNIL	OTH	7217	12218	12768
20	METOLACHLOR & METOLACHLOR-S	OTH	6171	6473	8121
21	METOLACHLOR-S	OTH	6171	6473	7951
22	PENDIMETHALIN	OTH	5863	6844	6693
23	COPPER	INO	5484	4197	3612
24	LINURON	OTH	5416	3900	2693
25	DIMETHENAMID & DIMETHENAMID-P	OTH	5318	4037	3724
26	DIMETHENAMID-P	OTH	5318	4037	3724
27	THIOPHANATE-METHYL	OTH	5204	9868	18533
28	DIMETHOATE	OP	4502	4110	4976
29	COPPER HYDROXIDE	INO	4420	4288	3375
30	DIQUAT	DIP	4403	3118	3232
31	MCPA	CPX	4084	7772	9301
32	MALEIC HYDRAZIDE	OTH	3644	4108	4439
33	COPPER OXYCHLORIDE	INO	3171	2071	1427
34	KAOLIN CLAY	INO	3052	17519	12426
35	FLUAZINAM	OTH	3017	2437	2521
36	BURKHOLDERIA SPP	MIC	2949	4570	3430
37	PENTHIOPYRAD	OTH	2893	1834	1450
38	PHOSPHORIC ACID	INO	2812	2465	6731
39	CLOPYRALID	OTH	2562	1887	2003
40	MEFENOXAM	OTH	2429	2204	1947
41	COPPER SULFATE	INO	2268	2012	1651
42	METHOMYL	CB	2244	4222	3208
43	BENTAZONE	OTH	2110	5834	9545
44	BACILLUS THURINGIENSIS	MIC	2067	1128	906
45	FLUTOLANIL	OTH	1972	897	1268
46	GLUFOSINATE	OTH	1802	1005	982

Table 2: Top 80 EPest-high Estimates, Adams County, WA, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) (*continued*)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
47	DICAMBA	OTH	1716	2436	2242
48	ATRAZINE	TRZ	1393	1551	1354
49	SPIROMESIFEN	OTH	1328	1221	1266
50	CHLORPYRIFOS	OP	1318	2262	2286
51	CYMOXANIL	OTH	1309	1168	867
52	PROPARGITE	OTH	1306	1366	1992
53	PYRACLOSTROBIN	OTH	1207	2090	1999
54	PYROXASULFONE	OTH	1088	882	685
55	AZOXYSTROBIN	OTH	995	2270	1894
56	BOSCALID	OTH	983	2552	3097
57	PYMETROZINE	OTH	967	584	664
58	POTASSIUM BICARBONATE	INO	965	927	647
59	SPIROTETRAMAT	OTH	894	734	781
60	ETHALFLURALIN	OTH	842	3417	7270
61	FLUROXYPYR	OTH	736	1396	1417
62	BIFENTHRIN	PYR	732	514	431
63	PROPOXYCARBAZONE	OTH	726	502	466
64	DIMETHOMORPH	OTH	686	412	330
65	MCPB	CPX	677	361	229
66	IMIDACLOPRID	OTH	643	530	858
67	THIAMETHOXAM	OTH	604	784	909
68	IMAZAMOX	OTH	603	510	556
69	TRIFLURALIN	OTH	603	871	690
70	ZOXAMIDE	OTH	580	283	542
71	DIAZINON	OP	578	203	539
72	CARBARYL	CB	520	963	1412
73	CLETHODIM	OTH	519	496	812
74	DIFENOCONAZOLE	OTH	461	292	332
75	TOLFENPYRAD	OTH	454	454	454
76	FAMOXADONE	OTH	443	746	556
77	FLUMIOXAZIN	OTH	425	319	223
78	IPRODIONE	OTH	424	2335	2491
79	MANDIPROPAMID	OTH	423	267	184
80	CHLORANTRANILIPROLE	OTH	405	332	341

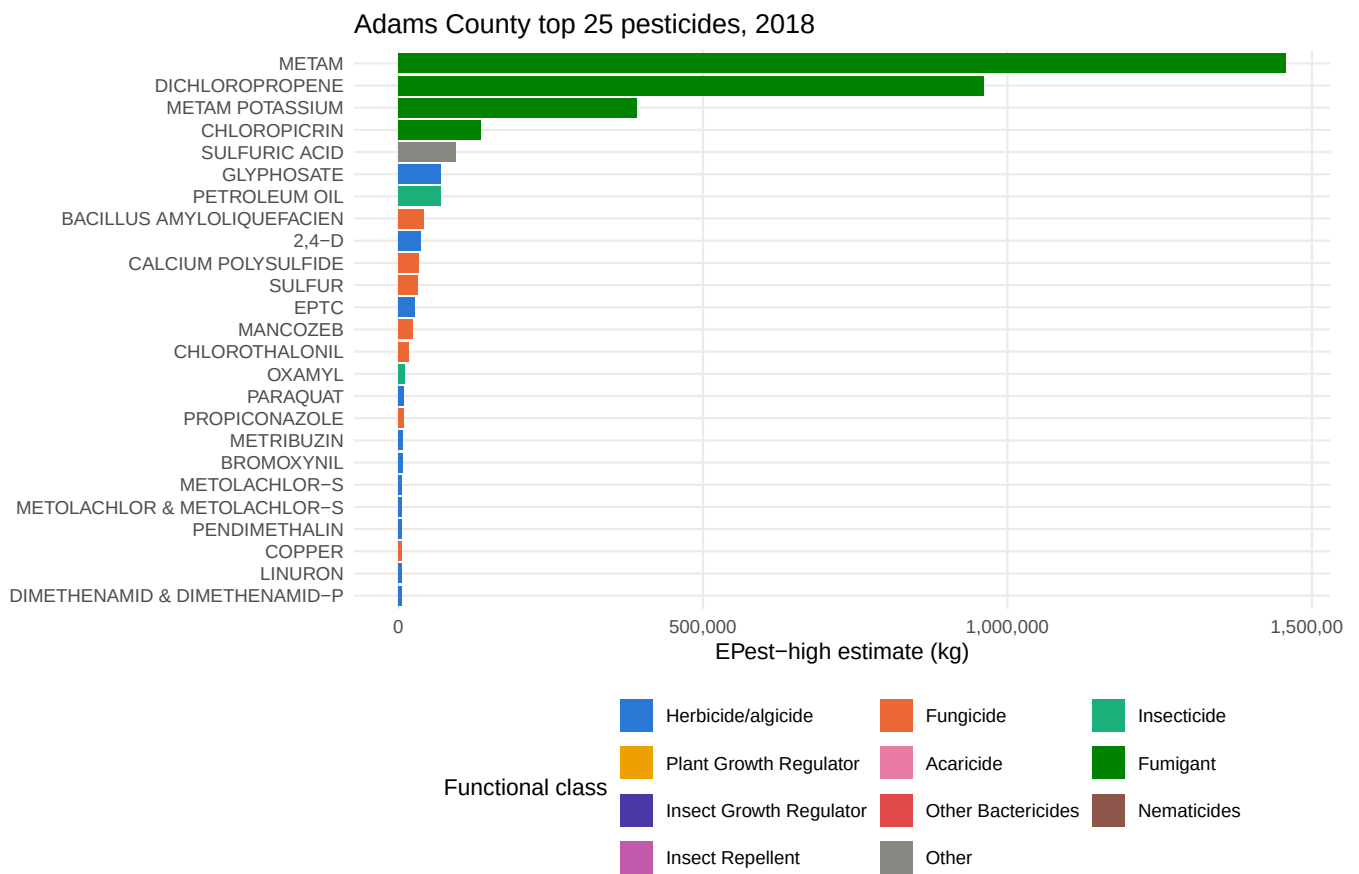


Figure 1: Top 25 compounds in Adams County by E Pest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Adams County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM	1457865	1457865
2	DICHLOROPROPENE	961753	961753
3	METAM POTASSIUM	392225	392225
4	CHLOROPICRIN	0	134727
5	SULFURIC ACID	0	94991
6	GLYPHOSATE	69962	70203
7	PETROLEUM OIL	66469	69034
8	BACILLUS AMYLOLIQUEFACIEN	41755	41755
9	2,4-D	37326	37473
10	CALCIUM POLYSULFIDE	32968	32979
11	SULFUR	32093	32328
12	EPTC	26346	26346
13	MANCOZEB	24417	24531
14	CHLOROTHALONIL	17753	17753
15	OXAMYL	10804	10804
16	PARAQUAT	8858	8913
17	PROPICONAZOLE	8292	8292
18	METRIBUZIN	3737	7945
19	BROMOXYNIL	7217	7217
20	METOLACHLOR & METOLACHLOR-S	6171	6171
21	METOLACHLOR-S	6171	6171
22	PENDIMETHALIN	5714	5863
23	COPPER	5322	5484
24	LINURON	5386	5416
25	DIMETHENAMID & DIMETHENAMID-P	4512	5318
26	DIMETHENAMID-P	4512	5318
27	THIOPHANATE-METHYL	5054	5204
28	DIMETHOATE	4502	4502
29	COPPER HYDROXIDE	4408	4420
30	DIQUAT	4403	4403
31	MCPA	4084	4084
32	MALEIC HYDRAZIDE	3644	3644
33	COPPER OXYCHLORIDE	3171	3171
34	KAOLIN CLAY	2897	3052
35	FLUAZINAM	3017	3017
36	BURKHOLDERIA SPP	2670	2949
37	PENTHIOPYRAD	2823	2893
38	PHOSPHORIC ACID	2812	2812
39	CLOPYRALID	47	2562
40	MEFENOXAM	2402	2429
41	COPPER SULFATE	2242	2268
42	METHOMYL	2244	2244
43	BENTAZONE	2110	2110
44	BACILLUS THURINGIENSIS	2064	2067
45	FLUTOLANIL	1972	1972
46	GLUFOSINATE	1802	1802
47	DICAMBA	1716	1716
48	ATRAZINE	1393	1393
49	SPIROMESIFEN	1328	1328
50	CHLORPYRIFOS	1279	1318

Table 4: Top 50 Adams County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	1428457	1428457
2	DICHLOROPROPENE	985065	985098
3	METAM POTASSIUM	257672	257672
4	SULFURIC ACID	0	208553
5	PETROLEUM OIL	91811	95154
6	GLYPHOSATE	91992	94352
7	CHLOROPICRIN	13351	58619
8	BACILLUS AMYLOLIQUEFACIEN	44305	44305
9	2,4-D	38351	38569
10	SULFUR	35159	35393
11	MANCOZEB	30887	30925
12	EPTC	30235	30235
13	CALCIUM POLYSULFIDE	20298	20302
14	CHLOROTHALONIL	17950	17950
15	KAOLIN CLAY	6245	17519
16	BROMOXYNIL	9725	12218
17	PARAQUAT	11013	11569
18	THIOPHANATE-METHYL	9732	9868
19	PROPICONAZOLE	8307	8488
20	MCPA	5250	7772
21	METRIBUZIN	5626	7420
22	PENDIMETHALIN	6372	6844
23	METOLACHLOR & METOLACHLOR-S	5147	6473
24	METOLACHLOR-S	5147	6473
25	TRI-ALLATE	0	6357
26	BENTAZONE	5834	5834
27	ETHOPROPHOS	5761	5761
28	BURKHOLDERIA SPP	4434	4570
29	OXAMYL	4136	4336
30	COPPER HYDROXIDE	4164	4288
31	METHOMYL	4222	4222
32	COPPER	4143	4197
33	DIMETHOATE	4104	4110
34	MALEIC HYDRAZIDE	4108	4108
35	DIMETHENAMID & DIMETHENAMID-P	3736	4037
36	DIMETHENAMID-P	3736	4037
37	LINURON	2737	3900
38	ETHALFLURALIN	3417	3417
39	CUPROUS OXIDE	3152	3173
40	DIQUAT	3118	3118
41	BOSCALID	2550	2552
42	PHOSPHORIC ACID	1571	2465
43	FLUAZINAM	2437	2437
44	DICAMBA	2075	2436
45	IPRODIONE	2335	2335
46	AZOXYSTROBIN	2130	2270
47	CHLORPYRIFOS	1739	2262
48	MEFENOXAM	2195	2204
49	ZIRAM	0	2118
50	PYRACLOSTROBIN	1678	2090

Table 5: Top 50 Adams County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	1133223	1133223
2	DICHLOROPROPENE	932449	932557
3	METAM POTASSIUM	263799	264637
4	SULFURIC ACID	0	224700
5	PETROLEUM OIL	91150	93156
6	GLYPHOSATE	84381	86057
7	EPTC	39169	39296
8	MANCOZEB	38624	38678
9	CHLOROPICRIN	10871	38585
10	SULFUR	38032	38421
11	2,4-D	36687	37654
12	BACILLUS AMYLOLIQUEFACIEN	31814	31814
13	THIOPHANATE-METHYL	18431	18533
14	CHLOROTHALONIL	18374	18375
15	CALCIUM POLYSULFIDE	17258	17394
16	BROMOXYNIL	10759	12768
17	KAOLIN CLAY	5662	12426
18	PARAQUAT	8819	9599
19	BENTAZONE	9545	9545
20	MCPA	7784	9301
21	OXAMYL	8127	8247
22	METOLACHLOR & METOLACHLOR-S	7292	8121
23	METOLACHLOR-S	7150	7951
24	METRIBUZIN	5906	7528
25	ETHALFLURALIN	7270	7270
26	PHOSPHORIC ACID	945	6731
27	PENDIMETHALIN	6316	6693
28	PROPICONAZOLE	6472	6581
29	ETHOPROPHOS	5183	5183
30	DIMETHOATE	4652	4976
31	MALEIC HYDRAZIDE	4439	4439
32	TRI-ALLATE	0	4064
33	DIMETHENAMID & DIMETHENAMID-P	3500	3724
34	DIMETHENAMID-P	3500	3724
35	COPPER	3546	3612
36	BURKHOLDERIA SPP	3326	3430
37	COPPER HYDROXIDE	3287	3375
38	DIQUAT	3232	3232
39	METHOMYL	3018	3208
40	BOSCALID	3090	3097
41	METIRAM	2210	2872
42	LINURON	1885	2693
43	CUPROUS OXIDE	2424	2580
44	FLUAZINAM	2521	2521
45	IPRODIONE	2490	2491
46	CHLORPYRIFOS	1878	2286
47	DICAMBA	1604	2242
48	CHENOPODIUM AMBROSIOIDES	0	2168
49	CLOPYRALID	20	2003
50	PYRACLOSTROBIN	1573	1999